

ing diagram tracer to analyse events in the logic networks and to interact step-by-step with the circuit) and a finite-state machine editor/simulator (the algorithm for which is described using the ASM graphical editor). It also contains a microcomputer board emulator (the board includes an 8-bit CPU, ROM, RAM and the requisite input/output ports) and an assembler/interactive debugger module. The schematic editor lets you connect traditional logic circuits with sub-systems with the help of a finite-state machine editor and a microcomputer emulator. Hence it is possible to experiment with systems controlled by state machines and microcomputers with port interfacing and low-level programming of embedded systems.

Digital circuit simulator d-DcS appears to the user as a graphical schematic editor with a library of simplified logic components customised to pedagogical needs and not describing specific commercial products. As mentioned earlier, the schematic editor allows you to build simple digital networks composed of gates, flip-flops, predefined combinational and sequential circuits, and custom-defined components, which are defined as finite-state machines. Simulation can be interactive or in timing-mode.

In interactive mode, the student can animate the digital system in the editor, thereby controlling its inputs and observing results. This is the simplest way to examine a digital network and hence highly useful for beginners.

In timing mode, the behaviour of the circuit can be analysed by a timing diagram window, in which the user can define graphically an input signal sequence and observe the simulation results. This mode is analogous to professional simulation.

Finite-state machine simulator d-FsM allows graphical editing and simulation of finite-state machine components using the algorithmic state machine paradigm. The tool allows local functional simulation of finite-

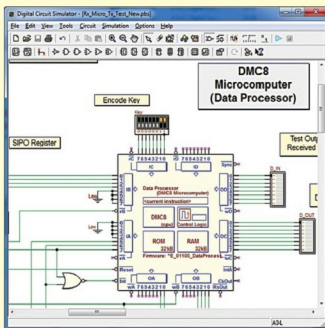


Fig. 2: An example of a circuit based on the microcomputer component

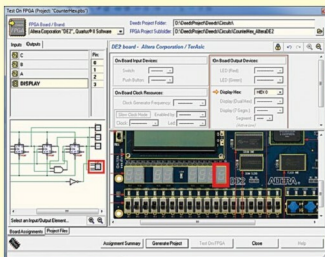


Fig. 3: View of the 'Test on FPGA' expert window, one of the new tools

state machines, which is user-designed with runtime display of relationships between state and timing evolution. Components produced by d-FsM can be directly used in d-DcS and inserted into any digital circuit. These can also be exported in VHDL language. A general-purpose finite-state machine software simulator helps students to enhance their design skills and also facilitates transition from pedagogical to professional field by introducing CAD methodologies.

In addition, the latest edition of the software offers users a new VHDL con-

verter (to export projects to other design environment tools) along with an FPGA 'expert' module to export and test projects on FPGA boards. The FPGA extension introduces FPGA-based systems for practice of digital design. New commands allow compilation of a project onto an FPGA chip by exporting it in VHDL to an FPGA-specific EDA tool. (The best part is that beginners to this concept can also utilise this feature as it doesn't require knowledge of VHDL language.)

With microcomputer emulator d-McE, users can practise programming at Assembly language level. It functionally emulates a board including a CPU, ROM and RAM, parallel

I/O ports, reset circuitry and a simple interrupt logic. The custom 8-bit CPU, named DMC8, is designed to suite your educational needs, and is based on a simplified version of the well-known Z80-CPU processor. The integrated source code editor enables users to enter Assembly programs, which can be assembled, linked and loaded in the emulated system memory with a simple command. The programs can be executed step by step in the interactive debugger. In the debugger, as in professional tools, the user can evaluate the contents of